

CAT 2025

Previous Year Paper

(Slot 2 – Quant)



1. If m and n are integers such that $(m + 2n)(2m + n) = 27$, then the maximum possible value of $2m - 3n$ is

Correct Answer : 17

Topic: Linear and Special Equations

2. If $\log_{64} x^2 + \log_8 \sqrt{y} + 3 \log_{512}(yz) = 4$, where x , y and z are positive real numbers, then the minimum possible value of $(x + y + z)$ is

- A. 96
- B. 48
- C. 36
- D. 24

Correct Answer : 48

Topic: Logarithms

3. The average number of copies of a book sold per day by a shopkeeper is 60 in the initial seven days and 63 in the initial eight days, after the book launch. On the ninth day, she sells 11 copies less than the eighth day, and the average number of copies sold per day from second day to ninth day becomes 66. The number of copies sold on the first day of the book launch is

Correct Answer : 49

Topic: Average, Mixture and Allegation

4. Let $f(x) = \frac{x}{(2x-1)}$ and $g(x) = \frac{x}{(x-1)}$. Then, the domain of the function $h(x) = f(g(x)) + g(f(x))$ is all real numbers except

- A. $-1, \frac{1}{2}$ and 1
- B. $\frac{1}{2}, 1$ and $\frac{3}{2}$
- C. $-\frac{1}{2}, \frac{1}{2}$ and 1
- D. $\frac{1}{2}$ and 1

Correct Answer : A

Topic: Functions and Graphs

5. If $9^{x^2+2x-3} - 4(3x^2 + 2x - 2) + 27 = 0$, then the product of all possible value of x is

- A. 5
- B. 20
- C. 30
- D. 15

Correct Answer : B

Topic: Quadratic Equations

6. Let ABCDEF be a regular hexagon and P and Q be the midpoints of AB and CD, respectively. Then, the ratio of the areas of trapezium PBCQ and hexagon ABCDEF is

- A. 5 : 24
- B. 7 : 24
- C. 6 : 25
- D. 6 : 19

Correct Answer : A

Topic: Mensuration

7. If a , b , c and d are integers such that their sum is 46, then the minimum possible value of $(a - b)^2 + (a - c)^2 + (a - d)^2$ is

Correct Answer : 2

Topic: Maxima Minima

8. Let a_n be the n^{th} term of a decreasing infinite geometric progression. If $a_1 + a_2 + a_3 = 52$ and $a_1a_2 + a_2a_3 + a_3a_1 = 624$, then the sum of this geometric progression is
- A. 63
 - B. 60
 - C. 54
 - D. 57

Correct Answer : C

Topic: Series and Sequences

9. Suppose a, b, c are three distinct natural numbers, such that $3ac = 8(a + b)$. Then, the smallest possible value of $3a + 2b + c$ is

Correct Answer : 12

Topic: Linear and Special Equations

10. Two tangents drawn from a point P touch a circle with center O at points Q and R . Points A and B lie on PQ and PR , respectively, such that AB is also tangent to the same circle. If $\angle AOB = 50^\circ$, then $\angle APB$, in degrees, equals

Correct Answer : 80

Topic: Geometry

11. The number of divisors of $(2^6 \times 3^5 \times 5^3 \times 7^2)$, which are of the form $(3r + 1)$, where r is a non-negative integer, is

- A. 42
- B. 36
- C. 56
- D. 24

Correct Answer : A

Topic: Number System and Modern Maths

12. Ankita is twice as efficient as Bipin, while Bipin is twice as efficient as Chandan. All three of them start together on a job, and Bipin leaves the job after 20 days. If the job got completed in 60 days, the number of days needed by Chandan to complete the job alone, is

Correct Answer : 340

Topic: Time Work Pipe & Cistern

13. The ratio of expenditures of Lakshmi and Meenakshi is $2 : 3$, and the ratio of income of Lakshmi to expenditure of Meenakshi is $6 : 7$. If excess of income over expenditure is saved by Lakshmi and Meenakshi, and the ratio of their savings is $4 : 9$, then the ratio of their incomes is

- A. $5 : 6$
- B. $3 : 5$
- C. $7 : 8$
- D. $2 : 1$

Correct Answer : B

Topic: Ratio Proportion and Variation

14. The set of all real values of x for which $(x^2 - |x + 9| + x) > 0$, is

- A. $(-\infty, -9) \cup (9, \infty)$
- B. $(-\infty, -3) \cup (3, \infty)$
- C. $(-9, -3) \cup (3, \infty)$
- D. $(-\infty, -9) \cup (3, \infty)$

Correct Answer : B

Topic: Inequalities and Modulus

15. An item with a cost price of Rs. 1650 is sold at a certain discount on a fixed marked price to earn a profit of 20% on the cost price. If the discount was doubled, the profit would have been Rs. 110. The rate of discount, in percentage, at which the profit percentage would be equal to the rate of discount, is nearest to

- A. 12
- B. 14
- C. 16
- D. 18

Correct Answer : B

Topic: Profit, Loss and Discount

16. The equation $3x^2 - 5x + p = 0$ and $2x^2 - 2x + q = 0$ have one common root. The sum of the other roots of these two equations is

- A. $\frac{2}{3} - p + \frac{3}{2}q$
- B. $\frac{8}{3} + p + \frac{1}{3}q$
- C. $\frac{2}{3} - 2p + \frac{2}{3}q$
- D. $\frac{8}{3} - p + \frac{3}{2}q$

Correct Answer : D

Topic: Quadratic Equations

17. A mixture of coffee and cocoa, 16% of which is coffee, costs Rs 240 per kg. Another mixture of coffee and cocoa, of which 36% is coffee, costs Rs 320 per kg. If a new mixture of coffee and cocoa costs Rs 376 per kg, then the quantity, in kg, of coffee in 10 kg of this new mixture is

- A. 4
- B. 6
- C. 2.5
- D. 5

Correct Answer : D

Topic: Average, Mixture and Allegation

18. A loan of Rs 1000 is fully repaid by two installments of Rs 530 and Rs 594, paid at the end of first and second year, respectively. If the interest is compounded annually, then the rate of interest, in percentage, is

- A. 8
- B. 10
- C. 11
- D. 9

Correct Answer : A

Topic: Simple Interest and Compound Interest

19. Rita and Sneha can row a boat at 5 km/h and 6 km/h in still water, respectively. In a river flowing with a constant velocity, Sneha takes 48 minutes more to row 14 km upstream than to row the same distance downstream. If Rita starts from a certain location in the river, and returns downstream to the same location, taking a total of 100 minutes, then the total distance, in km, Rita will cover is

Correct Answer : 8

Topic: Time, Speed and Distance

20. The sum of digits of the number $(625)^{65} \times (128)^{36}$, is

Correct Answer : 25

Topic: Number System and Modern Maths

21. In a $\triangle ABC$, points D and E are on the sides BC and AC, respectively. BE and AD intersect at point T such that $AT : AT = 4 : 3$ and $BE : BT = 5 : 4$. Point F lies on AC such that DF is parallel to BE. Then, $BD : CD$ is

- A. 9 : 4
- B. 15 : 4
- C. 11 : 4
- D. 7 : 4

Correct Answer : C

Topic: Geometry

22. A certain amount of money was divided among Pinu, Meena, Rinu and Seema. Pinu received 20% of the total amount and Meena received 40% of the remaining amount. If Seema received 20% less than Pinu, the ratio of the amounts received by Pinu and Rinu is

- A. 8 : 5
- B. 5 : 8
- C. 2 : 1
- D. 1 : 2

Correct Answer : B

Topic: Percentage